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ARTICLE INFO

Article history:

Received 20 May 2022

Revised 27 February 2023

Accepted 22 May 2023

Available online 08 June 2023

Keywords:

Video visual relation prediction

Dual attentional transformer

Video visual relation detection

ABSTRACT

Video visual relation detection (VidVRD) is to detect visual relations among instances as well as the trajectories of the corresponding subjects and objects in the video. Most current works improve the accuracy of tracking the objects but neglect the other key challenge, predicting the reliable visual relations in the videos, a vital meant for downstream tasks further. In this paper, we propose a dual attentional transformer network (VRD-DAT) for predicting the visual relations, also known as the predicates, in multi-relation videos. Specifically, our network first respectively targets modeling action visual predicates (Act-T) and spatial locating visual relations (Spa-T) via two parallel visual transformer structures simultaneously. Then, an attentional weighting module obtains the final precise merged visual relations. We conduct extensive experiments on two public datasets, ImageNet-VidVRD and VidOR, to demonstrate our model is capable of outperforming other state-of-the-art methods effectively on the task of video visual relation prediction. Quantitative and qualitative results also show that with more accurate visual relations, the performance of the video visual relation detection task can be further boosted.

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1. Introduction

With the rapid development and popularization of multimedia technology, computer network, and digital equipment, video data on the Internet is showing substantial growth [5,27,33]. Due to the requirement of individuals and enterprises to query and filter videos, automatic acquiring semantic information of videos has attracted more and more attention in computer vision [22,49,59]. Various tasks related to video understanding such as video object tracking [47,25], video action classification [2,3,54] have achieved great results. Researchers [45,52,6] begin to further study the diverse relations in videos which is more complex than the entity-level information of videos and name it as video visual relation detection (VidVRD) [45,13].

Video visual relation detection (VidVRD) aims to detect relation triplets presented as *(subject, predicate, object)* and trajectories of the corresponding subject and object in the video, which is a very

challenging goal. Existing works [45,13] mainly follow a pipeline as such: first, apply a video object detection model [41,10,57] to detect and generate the object tracklets on the decomposed overlapping video segments, and then classify each relational component to predict the short-term relation instances. After that, video visual relations can be generated through greedily associating the short-term relations. The prediction of the three relational components largely determines the final performance of this task. We focus on the problems encountered in the visual predicate prediction.

There are two challenges in the visual predicate prediction, which we try to overcome in this work. First, previous works [45,39,46,29,60] neglect the difference between action visual relations and spatial visual relations. As shown in Fig. 1, action visual predicate (or name as semantic visual predicate) contains “carry”, “watch”, “ride”, which all expresses action in videos. Spatial locating visual relation contains “in-front-of”, “behind”, “next-to”, etc., which all express the spatial relationship between objects. Previous works about action recognition [62,8] and spatial relation recognition [37,64] also demonstrate the difference between the two kinds of visual relations. Moreover, it can be seen that the action visual predicate only appears one time in each video in most cases, while the spatial locating relation tends to more, causing the action visual predicates usually to be ignored in prediction. There-

[☆] This work was supported by National Natural Science Funds of China (No. 62277011).

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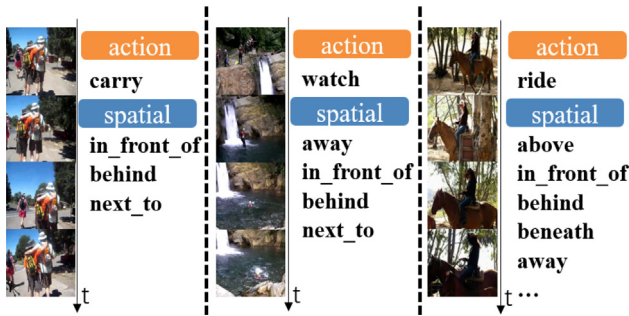


Fig. 1. Illustration of action visual predicates and spatial locating visual relations in each video on VidOR.

fore, we maintain that it is more appropriate to model and predict these two kinds of visual relations respectively by a specific visual predicate prediction model rather than a universal model, as the previous works did.

Second, current popular transformer-based model structures have achieved competitive results in the field of natural language processing (NLP) [12,11,40] and computer vision [66,3,15,31]. Transformer is now the dominant network for deep learning due to its advantages in parallel computing and processing long-term information. Video visual relation prediction also requires long-term information, but the performance of transformers on this task is still unexplored. Therefore, we investigate the potential ability of the visual transformers for visual relation prediction in our network.

To tackle these two challenges introduced above, we propose a network named “dual attentional transformer network (VRD-DAT)”, consisting of “Dual visual relation prediction transformer”, “combination module” and “attentional weighting”. To model multiple levels of the receptive field from the given video, we adopt Swin-Transformer [31] which contains the shifted-window structure. Specifically, the dual visual relation prediction transformer module consists of two parallel visual Swin-Transformer structures, namely action visual predicates transformer (Act-T) and spatial locating visual relations transformer (Spa-T). Unlike other dual attention networks, our dual attention module is divided according to the semantics of predicate categories and predicts action predicates and spatial predicates in videos in parallel. Then, the action and spatial combination module merges these two kinds of predicates into one candidate pool for preparing the next attentional weighting stage, since the division of action visual predicates and spatial locating visual relations is necessary and would boost the performance of most video analysis tasks [32,16,35]. After that, through attentional weighting, the final visual relations are generated from the relation predictions yielded by VidVRD-II [44] as well as our merged relation candidate pool.

The main contributions of this paper are summarized as follows:

1. We propose a dual attentional transformer network (VRD-DAT) for predicting the visual relations in multi-relation videos. We divide two attention modules according to the semantics of the predicate categories. It is capable of predicting visual predicates more precisely through separately modeling action visual predicates and spatial locating relations.
2. We first investigate the ability of the visual transformer-based model and demonstrate its advanced capabilities for video visual relation prediction in both quantitative and qualitative analysis.
3. We conduct extensive experiments on ImageNet-VidVRD and VidOR datasets. The proposed method consistently outperforms

state-of-the-art methods, improving the Precision@1 on the ImageNet-VidVRD by up to 3.60% on VidVRD-II [44] and 13.5% on VRD-STGC [30] in specific.

2. Related work

2.1. Video visual relation detection

Most of early researches focused on the study of the visual relation detection on images [65,36]. With the gradual maturity of image visual relation detection technology [31,63], more and more researchers begin to turn their attention to the video visual relation detection research. Compared with image visual relation detection (ImgVRD), video visual relation detection (VidVRD) is more difficult and complex. Most of the current research [6,13,46,48,61] follows Shang et al. [45] proposed three-stage detection framework, which is object tracklet proposal, relation prediction and greedy relational association. ImageNet-VidVRD dataset [45] and VidOR dataset [43] greatly encourages the research on video visual relation detection.

Recent studies in this area have shown good performance. For instance, Hsieh et al. [26] combines structured probabilistic models and deep networks in order to learn both the latent decomposition and disentanglement without explicit supervision. Sun et al. [48] used multi-model feature (language, spatial, temporal) fusion to improve prediction performance. Tsai et al. [52] proposed a novel gated energy parametrization function to learn adaptive relations conditioned on visual observations and amortize space complexity. Wu et al. [59] proposed a new network named ImagineRNN which uses comparative learning for optimization instead of feature regression. Gao et al. [23] designed a tracklet-based visual transformer for relation detection. Woo et al. [58] proposed a new method with advantages in efficiency and effectiveness, which need not resort to common external algorithms. Shang et al. [44] utilized an iterative manner to refine three components classification scores, which achieved great results on ImageNet-VidVRD. However, the problem of video visual predicate prediction has rarely been studied directly. We showed that it can improve the performance of the visual relation detection task effectively through using the precise visual predicates prediction.

2.2. Video vision transformers

At first, the video recognition task was basically dominated by convolution-based backbone architecture [18,38]. However, a few years ago, the emergence of the model consisting only of self-attention [53] changed this situation. This model abandoned the traditional convolution and recurrent network and achieved good results in the natural language processing (NLP). The excellent parallelism of transformer networks and the fact that the number of operations required to compute the association between two locations does not grow with distance has made it shine in the NLP. Inspired by Vaswani et al. [53], the model based on transformers has been used for various language tasks [9,12,11,40,4]. Based on the similarity in processing between videos and sentences, more and more researchers [56,3,51,1,17,31] have begun to use the transformer-based network in video recognition tasks. Arnab et al. [1] designed a pure-transformer architecture named ViViT which can achieve state-of-the-art results on multiple video classification benchmarks. Fan et al. [17] used the multi-scale vision transformer for training and achieved good results on SSv2 dataset [24]. Liu et al. [31] proposed a pure-transformer architecture based on the locality inductive bias, which can use strong pre-trained image models to improve recognition results. However, there is no previous research using transformer-based models for the visual relation prediction. The dual attention mechanism has also

been used for a variety of tasks. For instance, DaViT [14] utilized self-attention mechanisms with both “spatial tokens” and “channel tokens” for image tasks. Fu et al. [21] similarly designed spatial-based attention module and channel-based attention module for scene segmentation. Li et al. [28] used dual attention networks (spatial and channel) for hyperspectral image (HSI) classification in order to capture the large number of spectral and spatial features contained in the HSI. The dual attention mechanism we adopted is actually different from the division of most papers at present. The dual attention mechanism used in most papers is the spatial attention module and the channel attention module as described above. However, we divide the network according to the semantics of predicate categories, specifically the spatial locating attention module and the action attention module.

3. Method

In this section, we introduce our proposed dual attentional transformer network (VRD-DAT) for the video visual relation prediction task. Fig. 2 shows its specific structure, which is composed of three modules, namely “dual visual relation prediction transformer”, “combination module” and “attentional weighting”, respectively.

3.1. Dual visual relation prediction

3.1.1. Data partition

To facilitate the training of video classification models, we first extracted the video data according to the data annotation (predicate and corresponding start and end frames). We transformed each raw video into a label matrix $V, V \in \mathcal{V}, \mathcal{V} \in \mathbb{R}^{L \times M \times N}$, where $\mathcal{V}$ denotes the set of video label matrices; L denotes the total number of video data; M denotes the maximum number of the relations the video can contain; N denotes the number of video predicate categories. This label matrix set $\mathcal{V}$ confirms the one-to-one correspondence between videos and predicates, and we can perform the video extraction according to it.

This extracting method greatly increases the amount of the video data, and causes videos to repeat or overlap largely. One way to overcome this problem is to divide the video data. As we introduced before, the large proportion of spatial locating visual

relation in video data causes that action visual predicates are usually ignored in prediction. Therefore, we divided the dataset into two parts by action visual predicate categories $\mathcal{C}_s$ and spatial locating predicate categories $\mathcal{C}_l$. Note that the action visual predicate here refers to the predicate that appears less frequently in the video, and similarly the spatial locating visual predicate refers to the predicate that appears more frequently in the video. We determined which part a video should be classified into by calculating a marker matrix W , which is represented as Eq. 1.

$$\begin{cases} W_i^s = V_i \lambda_s^T, & V_i \in \mathcal{V} \\ W_i^l = V_i \lambda_l^T, & V_i \in \mathcal{V} \end{cases} \quad (1)$$

where $\lambda_s, \lambda_l \in \mathbb{R}^N$ denote the action visual predicate category vector and the spatial locating relation category vector, respectively, $W^s, W^l \in \mathbb{R}^{M \times N}$. The specific calculation of λ_s is shown as follows:

$$\begin{cases} \lambda_{s,i} = 1, & \text{if } i \in \mathcal{C}_s \\ \lambda_{s,i} = 0, & \text{if } i \notin \mathcal{C}_s \end{cases} \quad (2)$$

The calculation method of λ_l is the same. For the sake of convenience, we denoted the divided action visual predicate data part as $\mathcal{S}$, and the spatial locating predicate data part as $\mathcal{L}$. And then through using W^s and W^l we represented on Eq. 1, the $\mathcal{S}$ and $\mathcal{L}$ can be divided as follows:

$$\begin{cases} \mathcal{S} = \bigcup_{i=1}^L \left(\bigcup_{j=1}^M \{ \mathcal{V}_{ij} | \text{if } W_j^s > 0 \} \right) \\ \mathcal{L} = \bigcup_{i=1}^L \left(\bigcup_{j=1}^M \{ \mathcal{V}_{ij} | \text{if } W_j^l > 0 \} \right) \end{cases} \quad (3)$$

When $W_j^s > 0$, it means that the j -th segment of the original multi-relation video i is an action visual relation video segment, which should be divided into the action visual relation data part, W_j^l is the same. However, it is a little fuzzy for action and spatial partition on ImageNet-VidVRD [45]. It has more predicate categories and the predicate category boundaries are not clear such as “fly-behind” and “lie-next-to”. An example of the partition method on ImageNet-VidVRD is shown in Fig. 3. It can be seen that we put the predicate “ride” into the action part and others such as

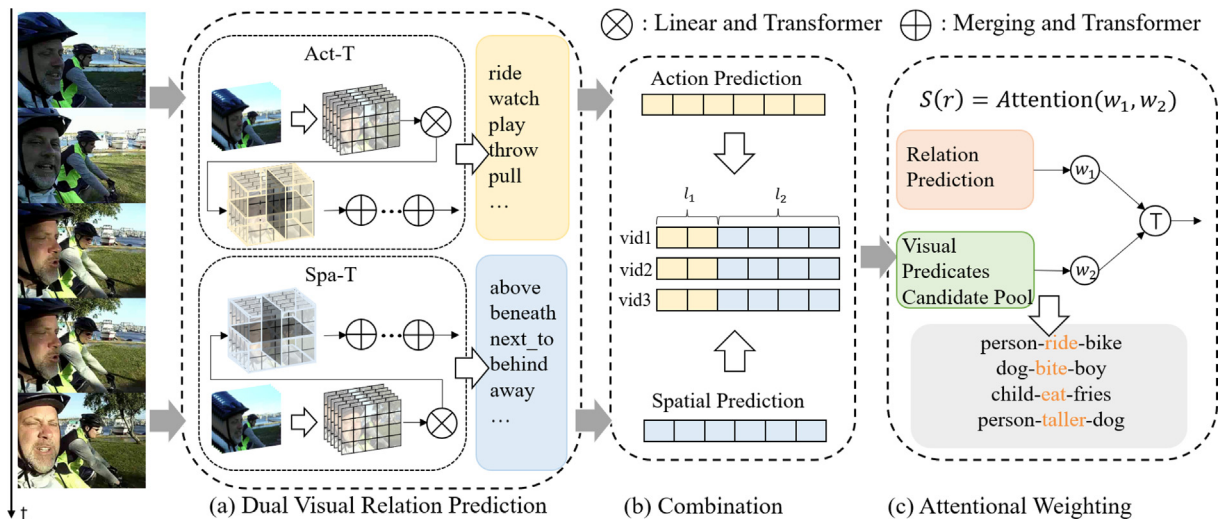


Fig. 2. Illustration of our proposed model VRD-DAT for Video Visual Relation Prediction. (a) The action visual predicates transformer (Act-T) and the spatial locating visual predicates transformer (Spa-T) respectively predict the corresponding result of predicates. (b) Obtain the visual predicates candidate pool through the combination block. (c) Through the attentional weighting module generate the final video visual relation predictions.

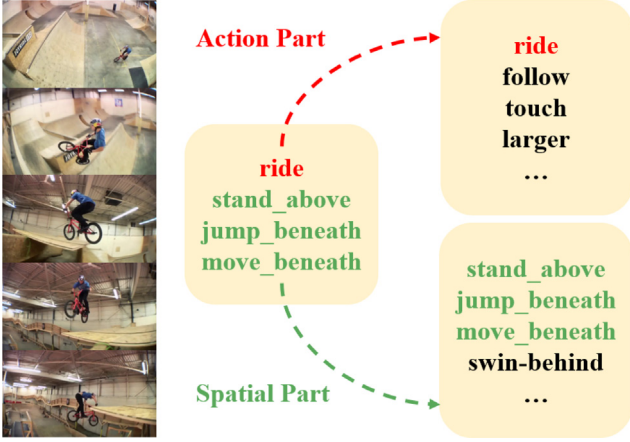


Fig. 3. Illustration of an action and spatial locating partition method example on ImageNet-VidVRD. The left column is the predicate set in the video, and the right column are the action and spatial locating sets we divided.

“stand-above” into the spatial locating part. Although this kind of predicates with the “action-location” form expresses both kinds of predicates at the same time, it can be observed from its frequency that it expresses more spatial locating information than the action. Such a partition method is convenient for us to set the combination number ratio α in the next module. Moreover, this method can greatly improve the prediction accuracy of the action visual predicate part and then improve the overall predicate prediction accuracy.

3.1.2. Predicate prediction

After data partition, the multi-relation videos are extracted and divided into the action visual predicate data part $\mathcal{S}$ and the spatial locating data part $\mathcal{L}$. And then we utilized two parallel relation prediction modules to train and predict the two data parts. To model multiple levels of the receptive field from the given video, we adopt Swin-Transformer [31] which contains the shifted-window structure as our single prediction module backbone. The graphical representation of prediction processes is shown in Fig. 2 (a). The input video is defined to be size of $T \times H \times W \times 3$, consisting of T frames which each contains $H \times W \times 3$ pixels. The first step in this process is 3D patch partition, each video is divided into $\frac{T}{2} \times \frac{H}{4} \times \frac{W}{4}$ 3D tokens (size $2 \times 4 \times 4 \times 3$), each patch/token consists of a 96-dimensional feature. After 3D patch partition, these 3D patches are computed by a linear embedding layer and a video Swin-Transformer block, which are annotated as “Linear and Transformer” in legend. The video Swin-Transformer block consists of a 3D shifted window based MSA module followed by a feed-forward natural network (2-layer MLP). The computational formulas of the two consecutive video Swin-Transformer blocks are shown in Eq. 4 [31].

$$\begin{aligned}\hat{f}^i &= \text{MSA}^{3D}(\text{LN}(f^{i-1})) + f^{i-1}, \\ f^i &= \text{FNN}(\text{LN}(\hat{f}^i)) + \hat{f}^i, \\ \hat{f}^{i+1} &= S - \text{MSA}^{3D}(\text{LN}(f^i)) + f^i, \\ f^{i+1} &= \text{FNN}(\text{LN}(\hat{f}^{i+1})) + \hat{f}^{i+1}\end{aligned}\quad (4)$$

where FNN denotes the feed-forward natural network; LN denotes the normalization layer; MSA^{3D} and $S - \text{MSA}^{3D}$ denote 3D window based multi-head self-attention (regular and shift window, respectively); $\hat{f}^i$ and f^i denote the output features of block i from MSA^{3D} ($S - \text{MSA}^{3D}$) module and FNN module, respectively. For computing the relative position bias, the calculation method of self-attention weights $\alpha_{(p,t)}$ for query patch (p, t) is shown as follows:

$$\alpha_{(p,t)} = \text{Softmax}\left(\frac{q_{(p,t)}^T}{\sqrt{D_h}} \cdot K_{(p,t)} + B\right) \quad (5)$$

where D_h denotes the variance of the dot product of query and key; p denotes the p -th patch and t denotes the index over frames; B denotes a small size bias matrix; K denotes the key associated with (p, t) . Finally, the computed tensors go through consecutive video Swin-Transformer blocks with patch merging modules, which are annotated as “Merging and Transformer” in legend and the prediction results can be generated. We used our trained action and spatial model to predict on original multi-relation videos for VidVRD. Through using the dual visual relation prediction module, the predicate classification performance is much better than using the direct transformers.

3.2. Action and spatial combination

After obtaining the predicate prediction results of the action part $\mathcal{S}$ and the spatial locating part $\mathcal{L}$, the results of two parts are merged to generate the final VidVRD relation candidate pool $\mathcal{R}$, the schematic diagram of our combination method is shown in Fig. 2 (b). Moreover, it can be represented as follows:

$$\mathcal{R}_i^v = P_i(\mathcal{S})\beta_{l_1} + P_i(\mathcal{L})\beta_{l_2} \quad (6)$$

where $\mathcal{R}_i^v$ denotes the relation candidate pool for the video i ; P_i denotes the predicate prediction results in descending order of probability for video i ; l_1, l_2 denote the number of action visual predicates and spatial locating predicates per video, respectively; β_{l_1}, β_{l_2} denote the label vectors computed by the l_1, l_2 , respectively. We defined the combination number ratio α as the number ratio of action visual predicates and spatial locating predicates in the relation candidate pool, so the l_1, l_2 can be computed like Eq. 7.

$$l_1 = \frac{p * \alpha}{(1 + \alpha)}, \quad l_2 = \frac{p}{(1 + \alpha)} \quad (7)$$

where p denotes the total number of generated predicates in the relation candidate pool. And then β_{l_1}, β_{l_2} can be computed as:

$$\beta_{l_1} = \gamma_{l_1}, \quad \beta_{l_2} = \beta_{l_1} \oplus \gamma_{l_1+l_2} \quad (8)$$

where γ_l denotes the column vector that the first l number is 1, and the remain number is all 0; $\beta_{l_1}, \beta_{l_2}, \gamma_{l_1}, \gamma_{l_2} \in \mathbb{R}^N$. The value of p depends on the number of relations in a video and the extent to which the final result is influenced by predicates. Too small p -value results in a relation prediction order that is not significantly different from before. Excessive p -value makes the relation prediction order depend too much on the predicates and ignore the influence of other relational components. The value of α depends on the proportion of action visual predicates to the overall predicates in the video. The larger the value of α , the more action visual predicates are predicted. We recommend setting p to 3 ~ 6 and α to 0.25 ~ 1.50. In our final experiment, we set p to 5 and α to 0.25.

3.3. Attentional weighting

This module is designed to generate the final relation predictions yielded by VidVRD-II [44] as well as our merged relation candidate pool. We employed an attention mechanism to weight the predicted relation triplets (VidVRD-II-based [44]) and our relation candidate pool, which can be computed as Eq. 9.

$$\text{Attention}((K, V), Q) = \frac{\exp(s(q, k_i))}{\sum_{j=1}^{\mathcal{S}} \exp(s(q, k_j))} v_i \quad (9)$$

where $(K, V) = [(k_1, v_1), (k_2, v_2), \dots, (k_{\mathcal{A}}, v_{\mathcal{A}})]$ denotes the key-value pair; Q denotes the query information; $s(q, k)$ denotes the attention scoring function, $q \in \mathbb{R}^Q$; $\mathcal{A}$ denotes the number of corresponding key-value pairs. For the design of the attention scoring function $s(q, k)$, since the query and key vector lengths are not the same, we adopt the additive attention scoring function, which is shown as Eq. 10:

$$s(q, k) = w_v^T \tanh(W_q q + W_k k) \in \mathbb{R} \quad (10)$$

where $W_q \in \mathbb{R}^{h \times q}$, $W_k \in \mathbb{R}^{h \times k}$, $w_v \in \mathbb{R}^h$ denote learnable parameters; h denotes the hyperparameter for the hidden unit.

Fig. 2 (c) sketches this attentional weighting method. w_1, w_2 are the weight adjustment factors for the visual relation prediction and the relation candidate pool. w_1 is used to adjust the weights of the original relational triples and w_2 is used to adjust the weights of the predicates. The larger w_1 is, the more the final visual relation prediction result is influenced by the initial visual relation prediction, and the larger w_2 is, the more the final visual relation prediction result is influenced by the result of the predicate prediction. Although our purpose is only to increase the appropriate confidence score to the relation triplets for which the predicate is correct, we have to consider dependencies on other components of the relation triplets. Obviously, the relation triplets with a higher confidence score are more likely to have the correct subject or object. Therefore, We added the weight adjustment factor w_1 for adjusting the weight for relation triplets. This attentional weighting method significantly improves the accuracy of action visual relation triplets, which can greatly improve the performance of video relation tagging task. Simultaneously, the performance of video relation detection task is also improved due to the addition of the weight adjustment factor w_1 for the relation triplets.

4. Experiment

4.1. Datasets

We adopted ImageNet-VidVRD dataset [45] and VidOR dataset [43] to perform our method. ImageNet-VidVRD dataset is the first dataset for the VidVRD task, which is extracted from ILSVRC2016-VID [42] with clear visual relation annotations. VidOR dataset is a large-scale dataset which contains 10 k videos (84 h) selected from YFCC-100 M [50]. Table 3 shows the statistical information of the two datasets. Compared with the ImageNet-VidVRD dataset, VidOR dataset has fewer predicate categories and more object categories. It has the 7,000: 835: 2,165 split of train-validation-test, and the test set in VidOR is not available, so we used the validation set for evaluation.

4.2. Evaluation protocol

For the VidVRD task, we also followed the evaluation protocol given by Shang et al. [45]. Considering the openness of visual object detection, Shang et al. [45] divided it into two different task evaluation metrics: relation detection and relation tagging. In the relation detection task, we call it correct if it contains correct relation triplets and its bounding box trajectories of subject and object have high enough vIoU compared to the trajectories in the ground truth. Same as the previous research [45,58], we set the overlapping threshold of vIoU to 0.5. Specifically, the relation detection task is evaluated by Mean Average Precision (mAP), Recall@50 (R@50) and Recall@100 (R@100). In the relation tagging task, it only needs to have the same relation triplets to be judged correct. This task removes the requirement for object detection and is suitable for some applications that only require relation triplets as a

result. Precision@1 (P@1), Precision@5 (P@5) and Precision (P@10) are used to evaluate this task.

Besides, we set evaluation metrics for the video multi-relation prediction. The evaluation metrics of this task need to measure the similarity between the predicted predicate set and the actual predicate set. We used Precision@5 (P@5) and Recall@5 (R@5) as the evaluation metrics. Accuracy@1 (Acc@1) and Accuracy@5 (Acc@5) are used to measure the results of single classification with our new extracted datasets. In order to rectify the problem of model overfitting, we employed a 5-fold cross-validation to obtain our optimal model parameters.

4.3. Comparison with state-of-the-art methods

The following recent state-of-the-art methods are compared:

1. VidVRD [45] is the first method for the VidVRD task and designed the three-stage model structure.
2. GSTEG [52] proposed a novel gated energy function parametrization that can improve computational efficiency and reduce space complexity.
3. MHA [46] generate more accurate relation triplets by maintaining multiple possible relation hypothesis.
4. MHRA [13] uses the generation of multiple hypotheses for video relation instances to enable more robust long-term relation prediction.
5. 3DRN [6] uses a novel deep network which can fuse the spatio-temporal visual characteristics, object label features, and spatial interactive features in order to obtain better performance.
6. VRD-GCN [39] achieves better predication results by the using of spatial-temporal contextual cues and an online association method.
7. VRD-STGC [30] uses a novel sliding-window scheme to simultaneously predict short-term and long-term relation instances.
8. TSPN [58] uses the full video context to simultaneously predict the temporal span and categories of the entire relations with lower time complexity.
9. VidVRD-II [44] utilizes a novel iterative relation inference approach to improve prediction results, which is also the base model we used to improve.

Table 1 and 2 show our comparison results on ImageNet-VidVRD dataset and VidOR dataset. It can be seen that our method surpasses the state-of-the-art methods on both datasets for video visual relation prediction. Results demonstrate that our improvement on ImageNet-VidVRD dataset is even higher, increasing P@1 in relation tagging by 3.60% on VidVRD-II [44]. This suggests that our method is more suitable for improvements on datasets with more predicate categories. Moreover, in terms of model complexity, our model is also simpler than VidVRD-II, and consumes less time after excluding the time we spend extracting the predicate videos from the dataset.

4.4. Implementation

As described in the Section 3, our model needs to perform predicate video extraction for each new dataset, which will consume some time, but it only needs to be performed once. According to the partition method of predicate categories we introduced, the predicate category ratio of ImageNet-VidVRD dataset is 106: 26, and VidOR dataset is 8: 42. The spatial locating part of VidOR dataset only has 8 predicate categories, but it also accounts for a large proportion of the dataset. After that, we divided the two datasets into action and spatial locating parts, respectively, according to

Table 1

Comparison of the state-of-the-art approaches for VidVRD on ImageNet-VidVRD dataset. The dataset was randomly divided into five disjoint data subsets for 5-fold cross-validation to obtain the optimal model parameters. For each of the 5 folds, we compute the performance of the proposed method on the testing data and compare it against VRD-DAT via a paired t-test to generate the p-values for the competitor method. ("†" denote that the significance level is reached as $p - value < 0.05$.)

Methods	Relation Detection			Relation Tagging		
	mAP	R@50	R@100	P@1	P@5	P@10
VidVRD	8.58	5.54	6.37	43.00	28.90	20.80
GSTEG	9.52	7.05	7.67	51.50	39.50	28.23
MHRA	13.27	6.82	7.39	41.00	28.70	20.95
3DRN	14.68	5.53	6.39	57.89	41.80	29.15
VRD-GCN	16.26	8.07	9.33	57.50	41.00	28.50
VRD-STGC	18.23	11.21	13.69	60.00	43.10	32.24
TSPN	18.90	11.56	14.13	60.50	43.80	33.73
VidVRD-II	29.37	19.63	22.92	70.40	53.88	40.16
VRD – DAT (std)	30.44[†] ±.0015	20.31[†] ±.0005	23.52[†] ±.0025	74.00[†] ±.0082	55.30[†] ±.0046	40.25[†] ±.0021

Table 2

Comparison of the state-of-the-art approaches for VidVRD on VidOR dataset. The dataset was randomly divided into five disjoint data subsets for 5-fold cross-validation to obtain the optimal model parameters. For each of the 5 folds, we compute the performance of the proposed method on the testing data and compare it against VRD-DAT via a paired t-test to generate the p-values for the competitor method. ("†" denote that the significance level is reached as $p - value < 0.05$.)

Methods	Relation Detection			Relation Tagging		
	mAP	R@50	R@100	P@1	P@5	P@10
3DRN	2.47	2.58	2.75	52.59	42.33	29.89
MHA	6.59	6.35	8.05	50.72	41.56	32.53
VRD-STGC	6.85	8.21	9.90	48.92	36.78	-
VidVRD-II	8.25	8.58	10.59	55.41	44.45	33.35
VRD – DAT (std)	8.29[†] ±.0015	8.59	10.61	55.89[†] ±.0047	44.50[†] ±.0008	33.39

Table 3

Comparison of statistical information of ImageNet-VidVRD dataset and VidOR dataset.

	ImageNet-VidVRD	VidOR
Total hours	3	84
Train videos	800	7,000
Val videos	-	835
Test videos	200	2,165
Object categories	35	80
Predicate categories	132	50

the predicate categories previously divided. The specific number of our divided datasets is illustrated as a bar chart in Fig. 5. It is shown that the split train:test of the spatial locating part and the action part on ImageNet-VidVRD dataset are 19,268: 6,649 and 3,972: 863, respectively. And on VidOR dataset are 205,451: 22,460 and 60,878: 7,106, respectively. Our statistical results hint that regardless of the dataset, the ratio of action visual predicates and spatial locating predicates is very different.

After data partition, we utilized the base version of video Swin-Transformer (layer numbers = 2, 2, 18, 2) with ImageNet-21 K pre-trained model for the single video classification module. Table 4

Table 4

Results of predicate prediction in ImageNet-VidVRD dataset and VidOR dataset.

ImageNet-VidVRD					
Settings	Acc@1	Acc@5		P@5	R@5
Action	0.24	0.71	Combination	0.55	0.44
Spatial	0.14	0.46			
VidOR					
Settings	Acc@1	Acc@5		P@5	R@5
Action	0.58	0.86	Combination	0.70	0.65
Spatial	0.38	0.95			

shows our predicate prediction results of the two datasets for VidVRD. It can be seen that the performance of the action part module is better than the spatial locating module. Moreover, Table 4 shows that the overall performance on VidOR dataset is better than the ImageNet-VidVRD. It is because the VidOR dataset has fewer predicate categories and is easier to distinguish between action visual predicates and spatial locating predicates. In terms of runtime, only the training of the visual predicate prediction model consumes some time (about a few hours), the time consumed in the subsequent predicate combination and attentional relation weighting is very short (about a few minutes).

4.5. Results analysis

Fig. 4 shows a comparison of before and after using our VRD-DAT model. Our model improves the performance by attentional weighting the relation instances whose predicate in our predicate prediction set. It can be seen that our model updated the correct relation instance “monkey-sit_right-monkey” and “red_panda-stand_right-red_panda” to the first position in the ranking, which can significantly improve P@1 value in video relation tagging task. The relation triplet “ monkey-sit_right-monkey” in the Fig. 4 left is not in the top three before, but through double weighted by rela-

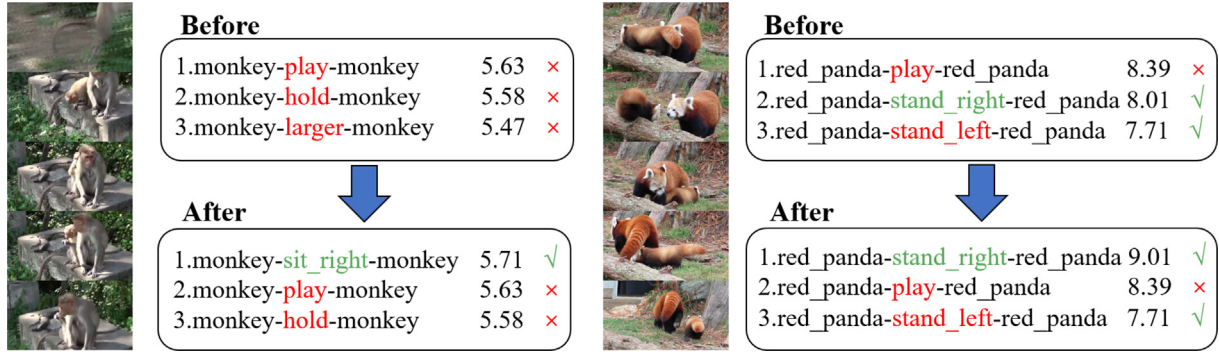


Fig. 4. Comparison of before and after using our dual attentional transformer network on ImageNet-VidVRD dataset. We extracted the first three predicted relation triplets and corresponding confidence scores for comparison. The predicates marked in green are within the relation candidate pool we predicted and in red are not in the candidate pool we predicted.

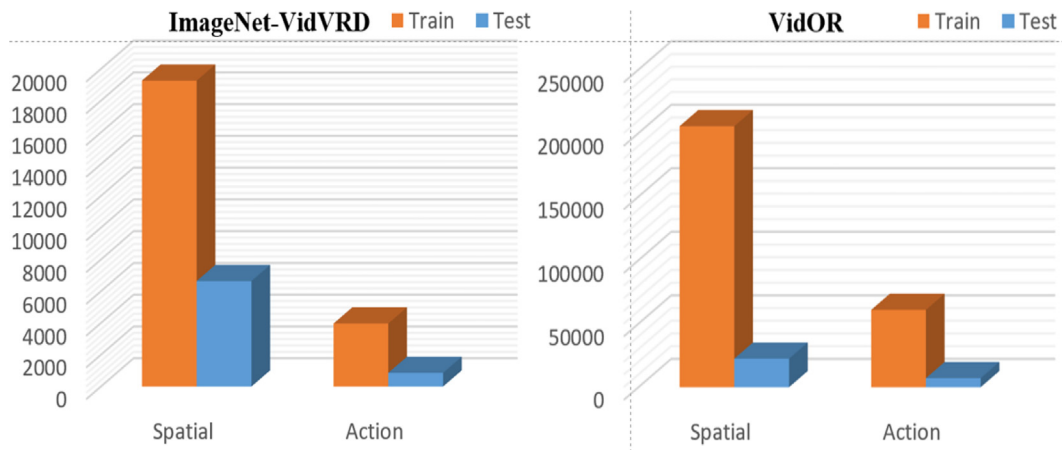


Fig. 5. Statistics histogram on extracted datasets derived from ImageNet-VidVRD and VidOR dataset. The ordinate is the number of videos.

tion prediction and relation candidate pool, its score increases rapidly. We did not lower the scores for relation instances whose predicates are not included in our relation candidate pool. It is because some relation triplets whose predicates are not in our relation candidate pool are also correct (examples in Fig. 4), and some multi-relation videos may have quite a few predicates, which is hard to predict thoroughly.

A comparison of using single and dual classification module on VidOR dataset is illustrated in Fig. 6. It is shown that using a single classification module for direct predicate prediction causes the lack of action visual predicates in prediction results. As discussed, this is due to the fact that the proportion of spatial locating visual relations in video data is too large. Putting action visual predicates into a module alone can overcome this problem very well. In addition, owing to the reduction of predicate categories for the classification module, the accuracy of the prediction is also improved. It can be seen that our model can predict an accurate action visual predicate. The prediction accuracy for spatial predicates is also improved (Fig. 6 left). This suggests that the performance of using the dual classification module (action and spatial locating) is significantly better than using the single classification module. However, our model also has some limitations. First, our model is more suitable for datasets with more predicate categories. Since our model is a dual prediction model based on predicates, it will have a weak improvement on the effect for datasets with few predicate categories. Second, our model prefers predicate categories that have a clear division for the number of occurrences in videos. That is, there is also a lack of improvement for datasets with significant

differences in the number of occurrences of each predicate category in the video. The resulting situation is that the ranking of the weights of the relations does not change after the initial prediction relation is weighted by the predicate.

4.6. Ablation study

We conduct extensive ablation studies for VRD-DAT model by showing the different classification modules, the different predicates combination parameters, and different weight adjustment factors, respectively illustrated in Section 3. Properly tuning the hyperparameters can make our model work better for a specific dataset. Obviously, if the parameters are not adjustable in the real world, we can also set the parameters to a universal value so that they can work well on different datasets. Note that we control other parameters consistent when conducting different settings for the investigated modules. The effectiveness investigations of each module are demonstrated by quantitative experiments furthermore.

4.6.1. Study on classification modules

We first utilized different classification module for direct predicate prediction on the subset of VidOR. The ratio of training set to test set is 34,446: 3,828, which is about 9: 1. The following classification methods are compared:

1. I3D [7] expands filters and pooling kernels of very deep image classification ConvNets into 3D, which can learn seamless

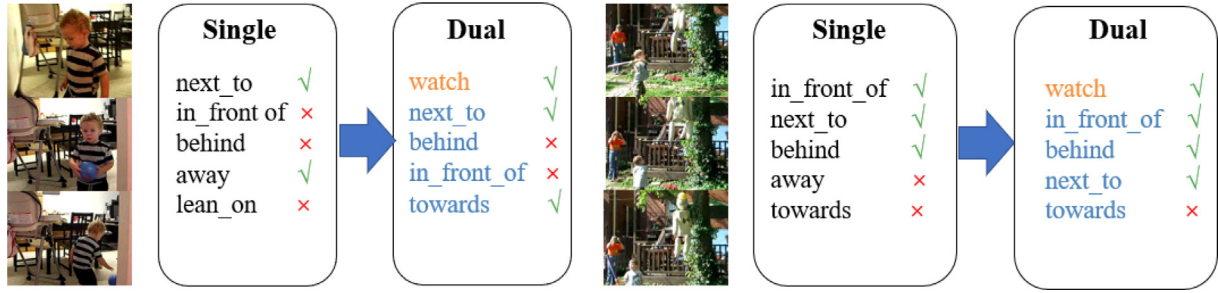


Fig. 6. Comparison of using single classification module and dual classification module (Act-T and Spa-T) on VidOR dataset. We extracted the top five predicates in relation candidate pool for comparison. The predicates marked in orange are predicted from the action part and blue are predicted from the spatial locating part.

Table 5

Comparison of different video classification modules in extracted subset of VidOR. All models were trained 100 epoches.

Methods	Acc@1	Acc@5
TimeSformer	0.31	0.79
I3D + NL	0.31	0.76
I3D	0.31	0.76
X3D	0.31	0.76
Swin-T	0.32	0.78
SlowFast	0.33	0.76

spatio-temporal feature extractors from video and use pre-trained models.

2. X3D [19] uses a simple stepwise network expansion approach which can achieve good accuracy to complexity trade-off.
3. Non-local operation [55] computes the response at a position as a weighted sum of the features at all positions, which can be used in many computer vision architectures.
4. TimeSformer [3] adapts the standard transformer architecture to video by enabling spatiotemporal feature learning directly from a sequence of frame-level patches.
5. SlowFast [20] used a slow pathway to capture spatial semantics, and a fast pathway to capture motion at fine temporal resolution.

The results can be seen in Table 5. It is shown that there is almost no difference in the direct classification performance of each method on the extracted dataset. It is because the video overlapping in the extracted dataset is too high, two almost identical videos may have different predicate categories, which limits the predictive power of various models. The results also show that using a single video classification module to predict visual predicates is poor. Therefore, we utilize I3D [7] and SlowFast [20] to train on our action data part and spatial locating data part. Results in Table 6 demonstrate that through partition for action and spatial locating part, the Accuracy@1 (Acc@1) and Accuracy@5 (Acc@5) significantly increased and the Swin-Transformer module has the best performance of Precision@5 (P@5) and Recall@5 (R@5).

Table 6

Results of dual video classification module in extracted subset of VidOR.

Methods	Settings	Acc@1	Acc@5	P@5	R@5
I3D	Action	0.43	0.71	0.63	0.58
	Spatial	0.39	0.92		
SlowFast	Action	0.45	0.74	0.64	0.59
	Spatial	0.41	0.94		
Swin-B	Action	0.49	0.77	0.69	0.62
	Spatial	0.36	0.92		

Table 7

Comparison of different values of parameter p for video visual relation on ImageNet-VidVRD dataset. To control the variables, we extracted an action visual predicate and $p - 1$ spatial locating visual predicates in the final predicate prediction.

p	P@1	P@5	P@10
4	73.20	55.20	40.20
5	74.00	55.30	40.25
6	72.80	55.10	40.10

Table 8

Comparison of different values of parameter α for video visual relation tagging on ImageNet-VidVRD dataset. The setting of parameter p is 5.

α	P@1	P@5	P@10
1.50	72.20	54.90	40.00
0.67	73.00	55.10	40.20
0.25	74.00	55.30	40.25

To prove the advantages of the transformer, we chose I3D [7] (CNN-based) and ViT-B-VTN (transformer-based) [34] as the classification module for video visual relation prediction on ImageNet-VidVRD dataset. Results are shown in Table 11. It demonstrates that the performance of transformer-based classification module is better than the CNN-based classification module.

4.6.2. Study on predicates combination

The number of predicates predicted in each video which is denoted as p in the relation candidate pool also affects the final result because predicting p predicates means attentional weighting p predicates. We recommend setting the p -value to the average of the total number of visual relations for each video. The comparison results of different values of p is shown in Table 7. It can be seen that the best results are obtained with $p = 5$. In the case of non-adjustable parameters, we recommend setting the p -value to 5 (based on experience with the two datasets). We defined the combination number ratio α as the ratio of the number of the action visual predicate part to the spatial locating relation part. An excessively large value of α will result in a disproportionately large share

Table 9

Results of different weight adjustment factors on ImageNet-VidVRD dataset. The dataset was randomly divided into five disjoint data subsets for 5-fold cross-validation to obtain the optimal model parameters.

Settings	mAP	Relation Detection R@50	R@100	P@1	Relation Tagging P@5	P@10
$w_1 = 0.3, w_2 = 0.3$	29.79	20.04	23.47	75.50	55.40	40.70
$w_1 = 0.1, w_2 = 0.5$	30.12	20.23	23.43	73.00	55.10	40.70
$w_1 = 0.05, w_2 = 0.2$	30.29	20.39	23.52	73.00	55.20	40.20
$w_1 = 0.1, w_2 = 0.2$	30.44	20.31	23.52	74.00	55.30	40.20

Table 10

Results of different weight adjustment factors on VidOR dataset. The dataset was randomly divided into five disjoint data subsets for 5-fold cross-validation to obtain the optimal model parameters.

Settings	mAP	Relation Detection R@50	R@100	P@1	Relation Tagging P@5	P@10
$w_1 = 0.1, w_2 = 0.05$	8.27	8.60	10.62	55.53	44.57	33.40
$w_1 = 0.1, w_2 = 0.2$	8.28	8.59	10.61	55.53	44.47	33.44
$w_1 = 0.1, w_2 = 0.2$	8.29	8.59	10.60	55.89	44.50	33.39

Table 11

Comparison of different classification module for video visual relation prediction on ImageNet-VidVRD dataset. The setting of weight adjustment factors is $w_1 = 0.1, w_2 = 0.2$.

Methods	mAP	Relation Detection		Relation Tagging		
		R@50	R@100	P@1	P@5	P@10
I3D	29.85	20.30	23.52	71.00	54.60	40.20
ViT	30.21	20.31	23.52	72.00	54.80	40.10
Swin	30.44	20.31	23.52	74.00	55.30	40.20

Table 12

Results of the single attentional weighting for action visual predicates prediction and spatial locating relations prediction on ImageNet-VidVRD dataset. "Act", "Spa" are used to represent the action data part and the spatial locating data part, respectively. The setting of weight adjustment factors is $w_1 = 0.1, w_2 = 0.2$.

Settings	mAP	Relation Detection		Relation Tagging		
		R@50	R@100	P@1	P@5	P@10
Act	29.74	20.35	23.45	72.50	54.80	40.10
Spa	30.34	20.35	23.54	70.50	55.20	40.30

Table 13

Results of video visual relation prediction without weighting relation prediction ($w_1 = 0$) on ImageNet-VidVRD dataset.

w_2	mAP	Relation Detection		Relation Tagging		
		R@50	R@100	P@1	P@5	P@10
0.1	30.03	20.39	23.37	71.50	54.90	40.05
0.2	30.16	20.41	23.54	72.00	55.00	40.10
0.3	30.21	20.29	23.50	72.50	54.80	40.15

of action visual predicates in the prediction results. Since the action visual predicates appear only 1–2 times in each video, we set its number to 1 and the spatial locating relations to 4 ($\alpha = 0.25$). In the case of non-adjustable parameters, we recommend setting the α -value to 0.25. The comparison results of different values of α is shown in Table 8. It can be seen that the relation tagging results for $\alpha = 0.25$ are better than those for $\alpha = 0.67$ and $\alpha = 1.5$. This study supports our previous definitions of the action part and the spatial part.

4.6.3. Study on weight adjustment factors

The setting of the weight adjustment factors plays an important role in our method. We performed a single weight test on action visual predicate results and spatial locating relation results

through setting one of the weights to 0. Results in Table 12 demonstrate that both modules have a certain improvement effect on the final results, but it is not as good as weighting at the same time. Moreover, we set w_1 to 0 to observe the effect on the relation prediction (VidVRD-II-based), which can be seen in Table 13. It is shown that without attentional weighting for relation prediction, the performance raises less than attentional weighting both the relation prediction and the relation candidate pool.

We show the results of our parameter tuning on ImageNet-VidVRD dataset in Table 9 and VidOR dataset in Table 10. We finally chose the $w_1 = 0.1, w_2 = 0.2$ set of experimental results for two datasets as our final results. As shown in tables, different weight adjustment factor settings have a considerable impact on the results. We found a set of interesting results in Table 9, that

is the result set with $w_1 = 0.3, w_2 = 0.3$. The relation tagging results of this group are all the best, but the relation detection results are very poor. It also enlightens us that in some cases where object detection and tracking are not required, such as video retrieval, we can obtain better prediction results by adjusting weighting adjustment factors. Obviously, in the case of non-adjustable parameters, we can set the parameters to $w_1 = 0.1, w_2 = 0.2$.

Moreover, the improvement effect of our model in ImageNet-VidVRD dataset is better than that in VidOR dataset. It is because we mainly improve the correctness of the overall relation instances by increasing predicate accuracy. As we introduced before, the VidOR dataset has fewer predicate categories and videos mostly focus on eight spatial predicate categories, which makes our model difficult to improve.

5. Conclusion

In this paper, we propose a dual attentional transformer network named VRD-DAT which first applies the transformer-based structures to the video visual relation prediction. Specifically, our network first respectively targets modeling action visual predicates (Act-T) and spatial locating visual relations (Spa-T) via two parallel visual transformer structures simultaneously. Then the action and spatial combination module merges the two kinds of predicates into one relation candidate pool. After that, the final visual relations are generated through an attentional weighting module. Extensive experiments on ImageNet-VidVRD and VidOR dataset show that our method achieves superior prediction results compared with the state-of-the-art methods. We also showed the advanced capabilities of the visual transformer-based model for video visual relation prediction. We will conduct more research on the transformer structure and make better use of it in various tasks of computer vision in the future.

CRedit authorship contribution statement

Mingcheng Qu: Conceptualization, Investigation. **Ganlin Deng:** Data curation, Writing - original draft. **Donglin Di:** Methodology, Writing - review & editing. **Jianxun Cui:** Supervision. **Tonghua Su:** Software, Validation.

Data availability

We conduct experiments on the public datasets.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tonghua Su reports financial support was provided by National Natural Science Funds of China.

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